GPU programming

Introduction

Alexey A. Romanenko
arom@nsu.ru
Novosibirsk State University

Computer power

«Titan» **2012**
299 008 Opteron Cores
18 688 k20 NVIDIA GPU
20+ Pflops



«K computer» **2011**
68 544 x 8-core SPARC64 VIIIfx.
8,162 Pflops











280 Tflops
212,992 CPUs

Performance

Time

Growth of performance

* Growth of clock rate
* Number of cores/CPUs
* CPU architecture complexity
  * Number of registers
  * Pipe-line
  * Digital capacity (4bit CPU, 16, 32, 64)
  * etc

Computer graphics





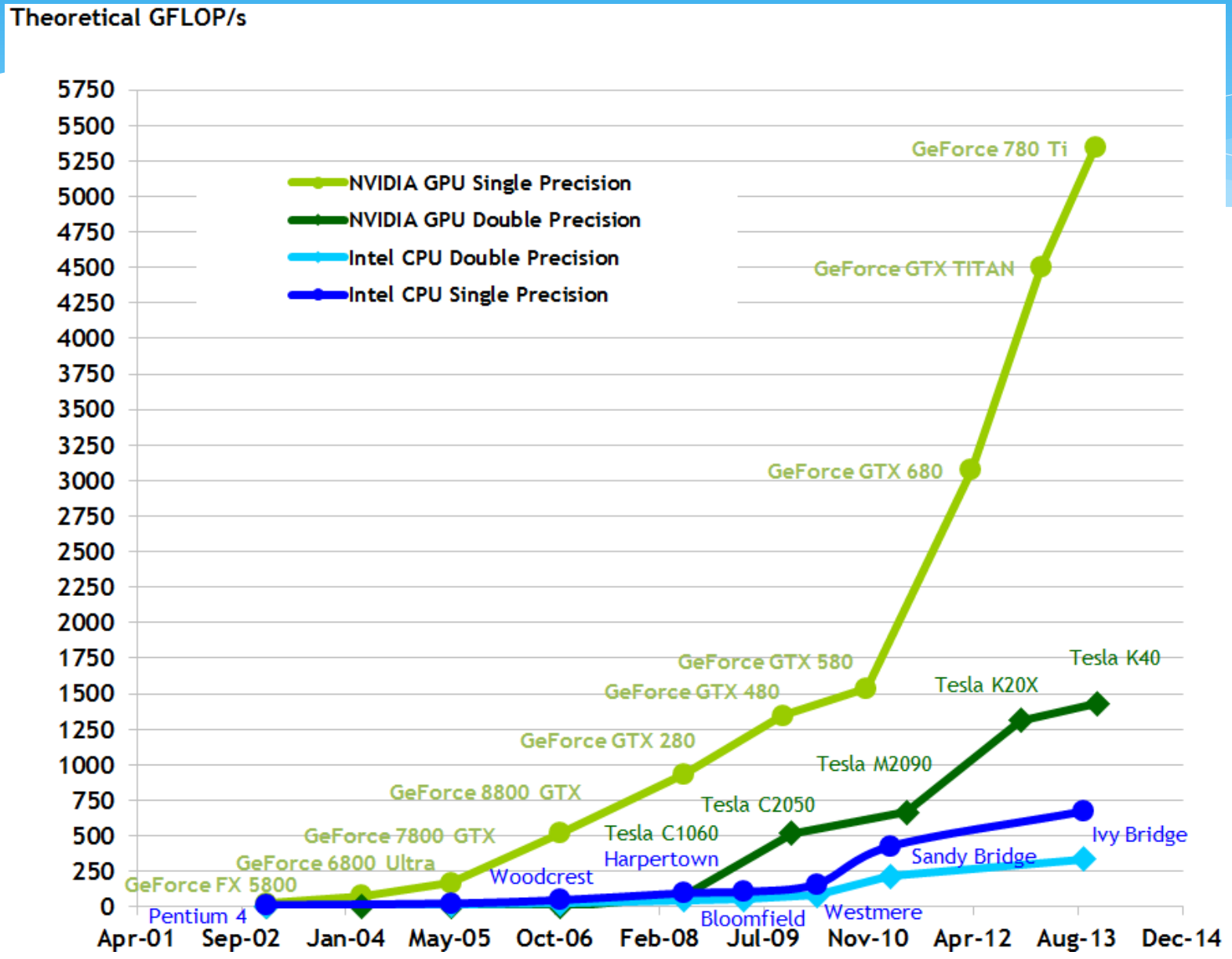
Computer graphics

* Small vectors
* Small matrixes
* filters/post-processing
* Calculation of projections
* etc.

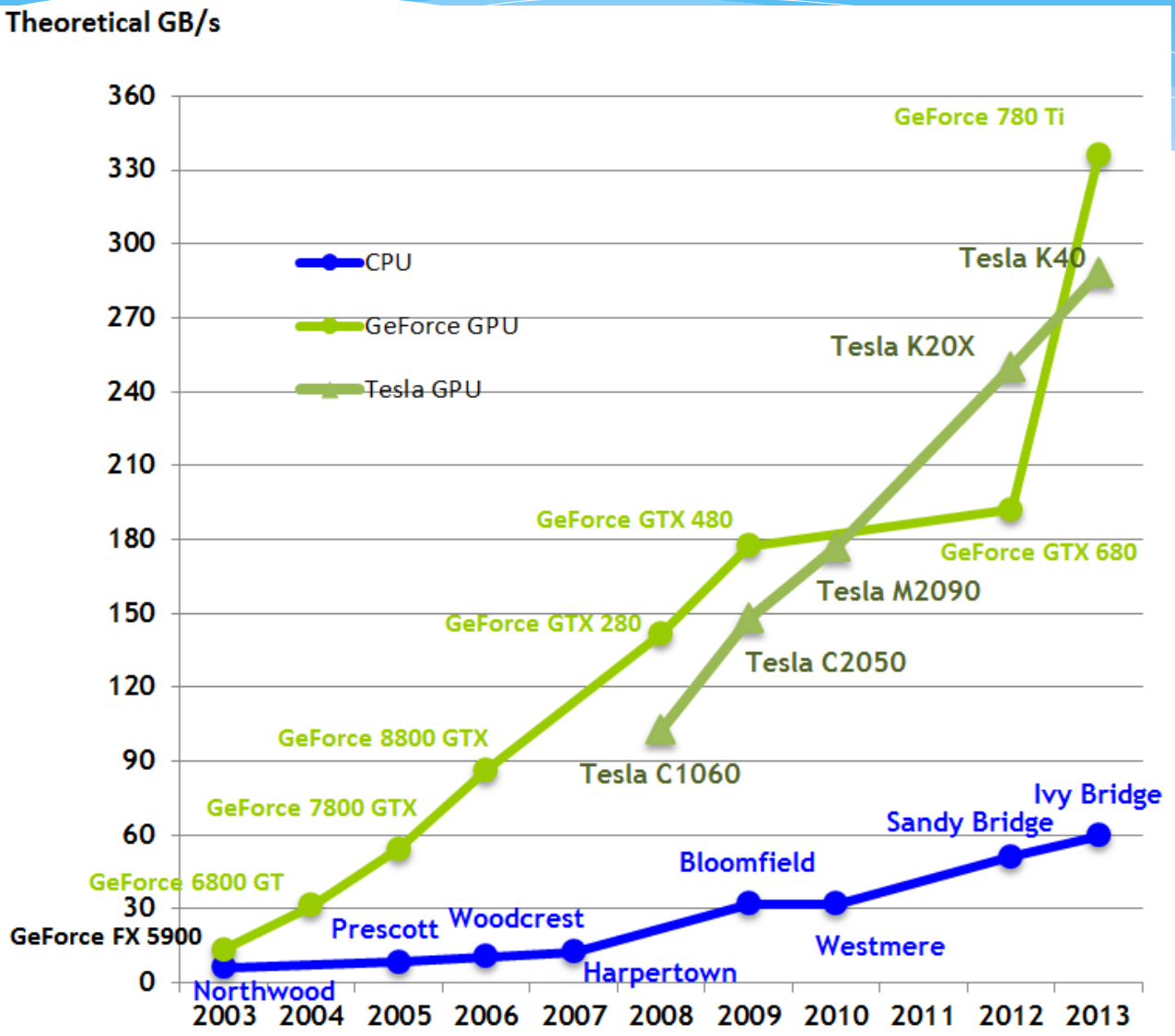


* Huge mumber of identical small operations.

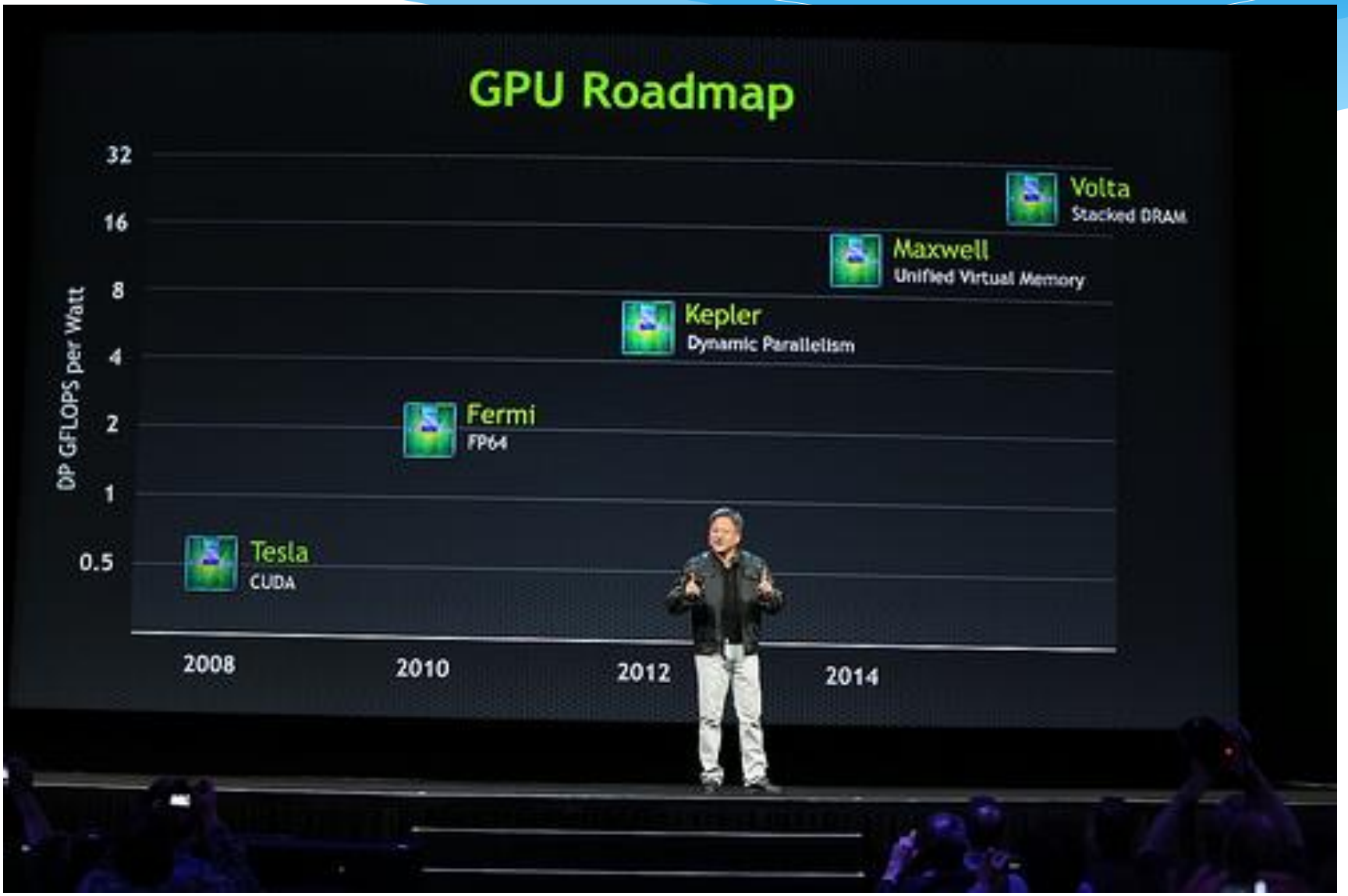
Performance of video cards



Memory bandwidth



March 2013



* Kepler: 28nm, 1.4 Tflops DP
* Maxwell: 22nm, 4 Tflops DP

Heterogeneous systems



+



9 of 10 leaders in top500 list!

Heterogeneous vs. Homogeneous computing systems (06.2012)

* **Tianhe-1A** (2 place)
  * Xeon X5670 6C 2.93 GHz, NVIDIA 2050



  * 4,7 Pflops (peak)
  * 4040 kW

* **Jaguar** (3 place)
  * Opteron 6-core 2.6 GHz



  * 2,3 Pflops (peak)
  * 6950 kW

November 2021. Top500 list

Rank	System	Cores	Rmax (TFlop/s)	Rpeak (TFlop/s)	Power (kW)
1	**Supercomputer Fugaku** - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442,010.0	537,212.0	29,899
2	**Summit** - IBM Power System AC922, IBM POWER9 22C 3.07GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband, IBM DOE/SC/Oak Ridge National Laboratory United States	2,414,592	148,600.0	200,794.9	10,096
3	**Sierra** - IBM Power System AC922, IBM POWER9 22C 3.1GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband, IBM / NVIDIA / Mellanox DOE/NNSA/LLNL United States	1,572,480	94,640.0	125,712.0	7,438
4	**Sunway TaihuLight** - Sunway MPP, Sunway SW26010 260C 1.45GHz, Sunway, NRCPC National Supercomputing Center in Wuxi China	10,649,600	93,014.6	125,435.9	15,371

Toward exaflop

Direct scaling…

- **Tianhe-1A**
- 4,7 Pflops (peak) * 213 = 1001,1 Pflops
- 4040 kW * 213 = 860 MW

- **Jaguar**
- 2,3 Pflops (peak) * 435 = 1000,5 Pflops
- 6950 kW * 435 = 3023 MW

exa	10^{18}
peta	10^{15}
tera	10^{12}
giga	10^{9}
mega	10^{6}
kilo	10^{3}

Sayano–Shushenskaya Dam



Turbines	10 × 640 MW (initial) 6 × 640 MW (current)
Installed capacity	5,120 MW (current)
Maximum capacity	6,400 MW
Annual generation	23.5 TW

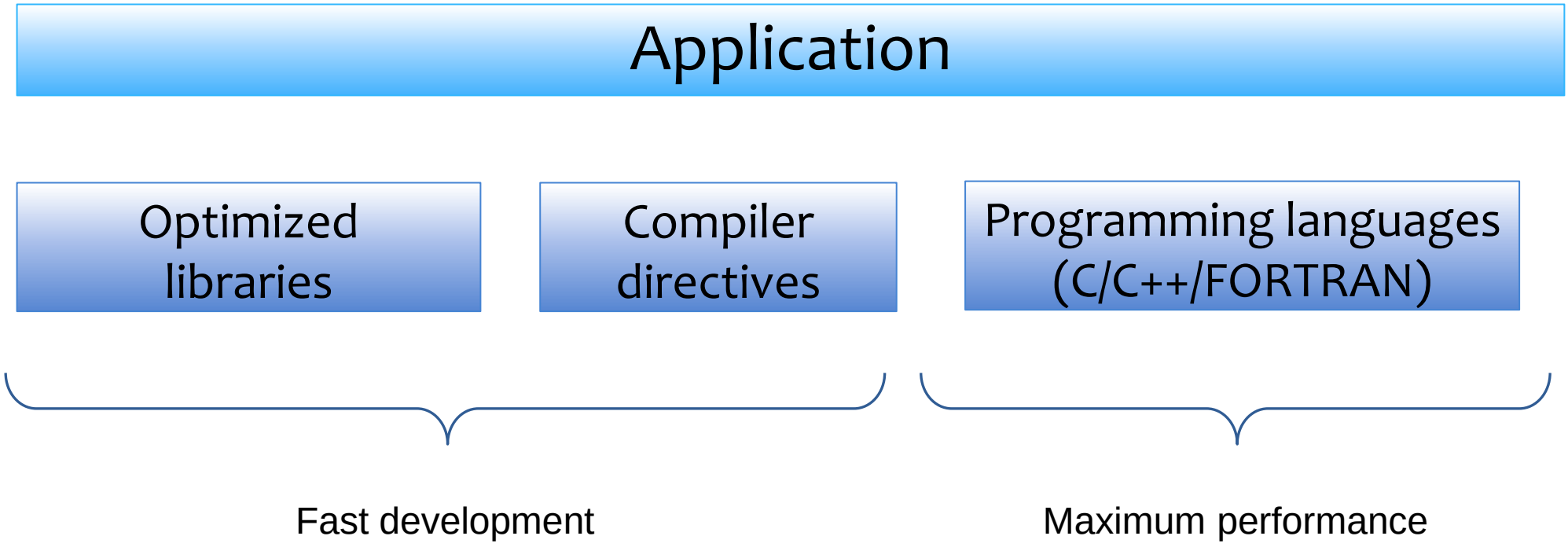
GFLOPs per Watt

* Green500 (November 2021)
  * https://top500.org/lists/green500
  * List of most power effective supercomputer
  * 9 of 10 from the top have NVIDIA GPUs.
* Performance / Watt
  * 39 Gflops/watt – now
  * 0,3 Gflops/watt– November, 2007
* Reaching exascale means boosting speeds by 50-100 times, while keeping power relatively static.

Application speedup

Example Applications	URL	Application Speedup
Seismic Database	http://www.headwave.com	66x to 100x
Mobile Phone Antenna Simulation	http://www.acceleware.com	45x
Molecular Dynamics	http://www.ks.uiuc.edu/Research/vmd	21x to 100x
Neuron Simulation	http://www.evolvedmachines.com	100x
MRI processing	http://bic-test.beckman.uiuc.edu	245x to 415x
Atmospheric Cloud Simulation	http://www.cs.clemson.edu/~jesteel/clouds.html	50x

Approaches to GPU programming



CUDA Roadmap



2022 – 15 years!

* CUDA 1.0 – 2007
* CUDA 2.0 – 2008
* CUDA 3.0 – 2009
* CUDA 4.0 – 2011
* CUDA 5.0 – 2012
* CUDA 5.5 – 2013

  …
* CUDA 11.5 - 2021

CUDA vs. OpenCL

* CUDA
  * NVIDIA GPU (Cray, HP, IBM, T-Platforms, NextIO…)
  * Close to peak performance
  * Functionality
  * Comfort for developers (debugger, performance analyzer, etc.)
  * Support
  * Teaching materials, libraries
* OpenCL
  * Architecture is not fixed, universality
  * Performance is not high priority

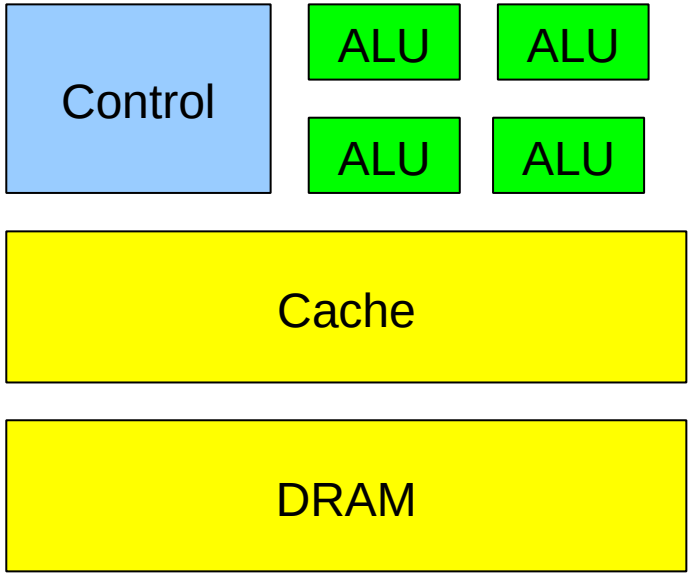
CUDA vs. OpenCL performance

* CUDA applications have up to 30% higher performance than OpenCL application.
  * http://arxiv.org/ftp/arxiv/papers/1005/1005.2581.pdf
  * http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=6047190

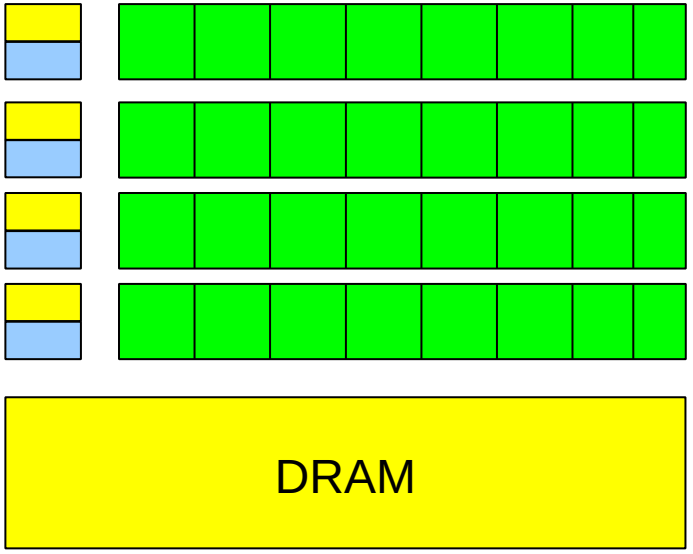
GPU — Graphical Processing Unit

* GPU is a CPU on video-card. GPU has its own parallel architecture (SIMD)
* Program on GPU can't communicate with the host
* Program on GPU can't update host memory
* Uploading/downloading data to/from GPU is performed via PCI Express. Host initiate data exchange process.
* GPU is a co-processor for CPU

CPU vs. GPU



CPU

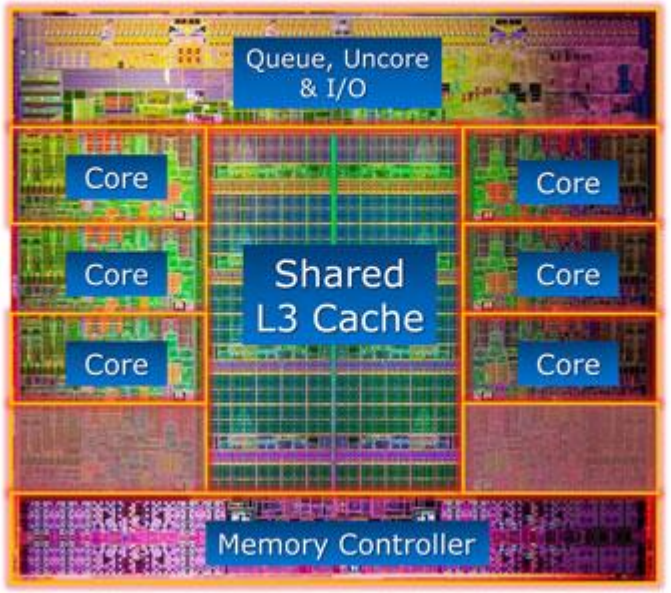


GPU

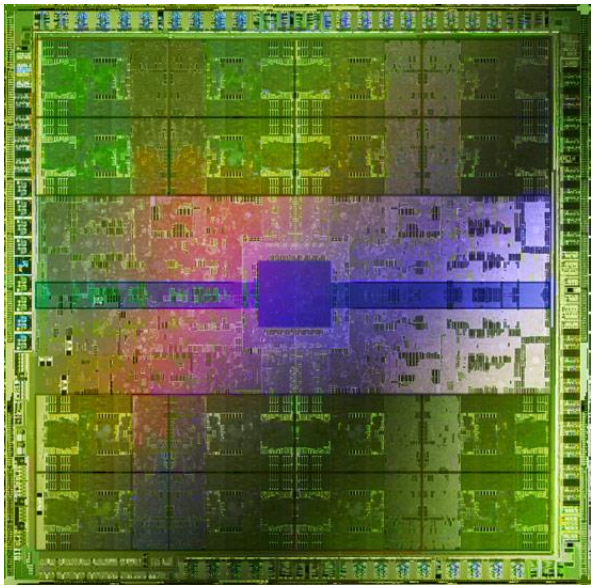
* Less die space for control and cache
* More die space for ALU

CPU vs. GPU

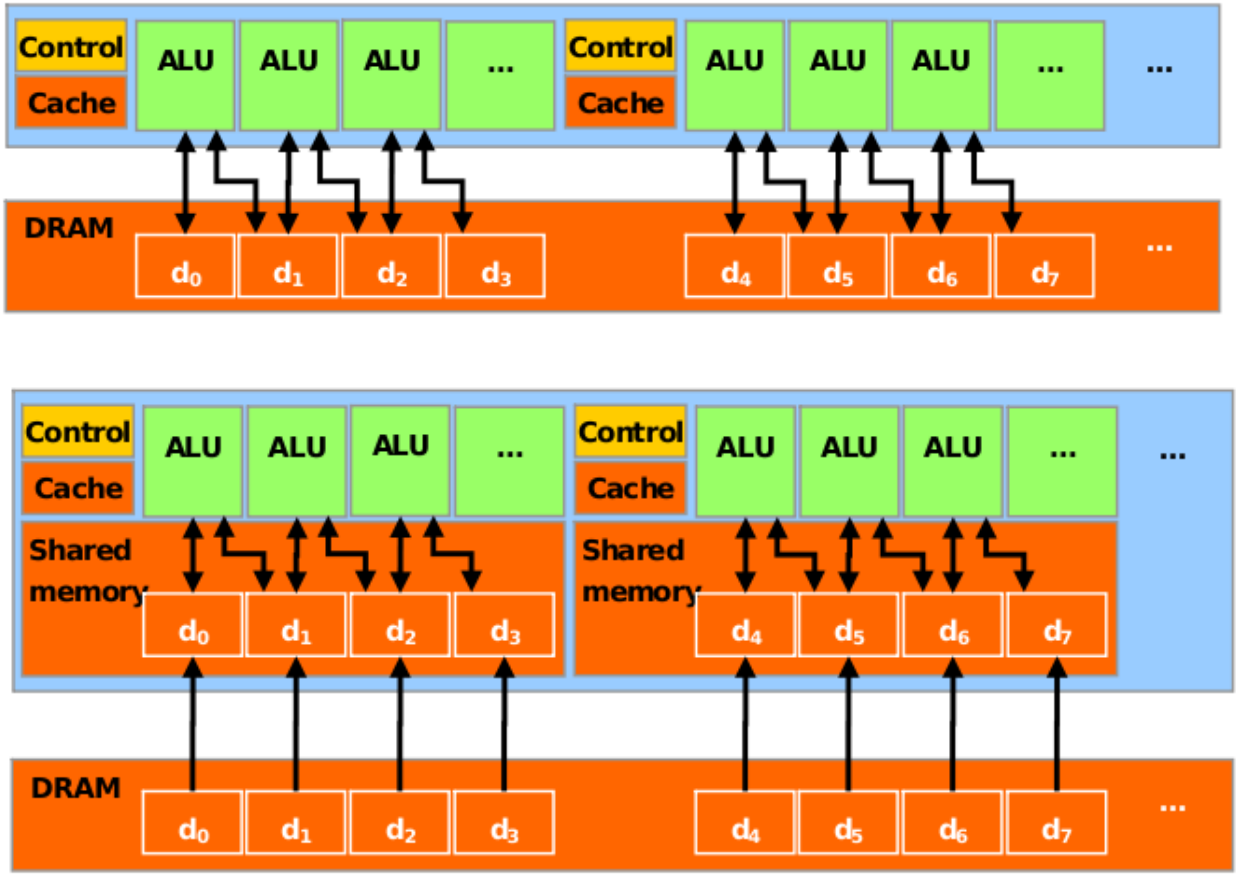
Intel® Core™ i7-3960X Processor Die Detail



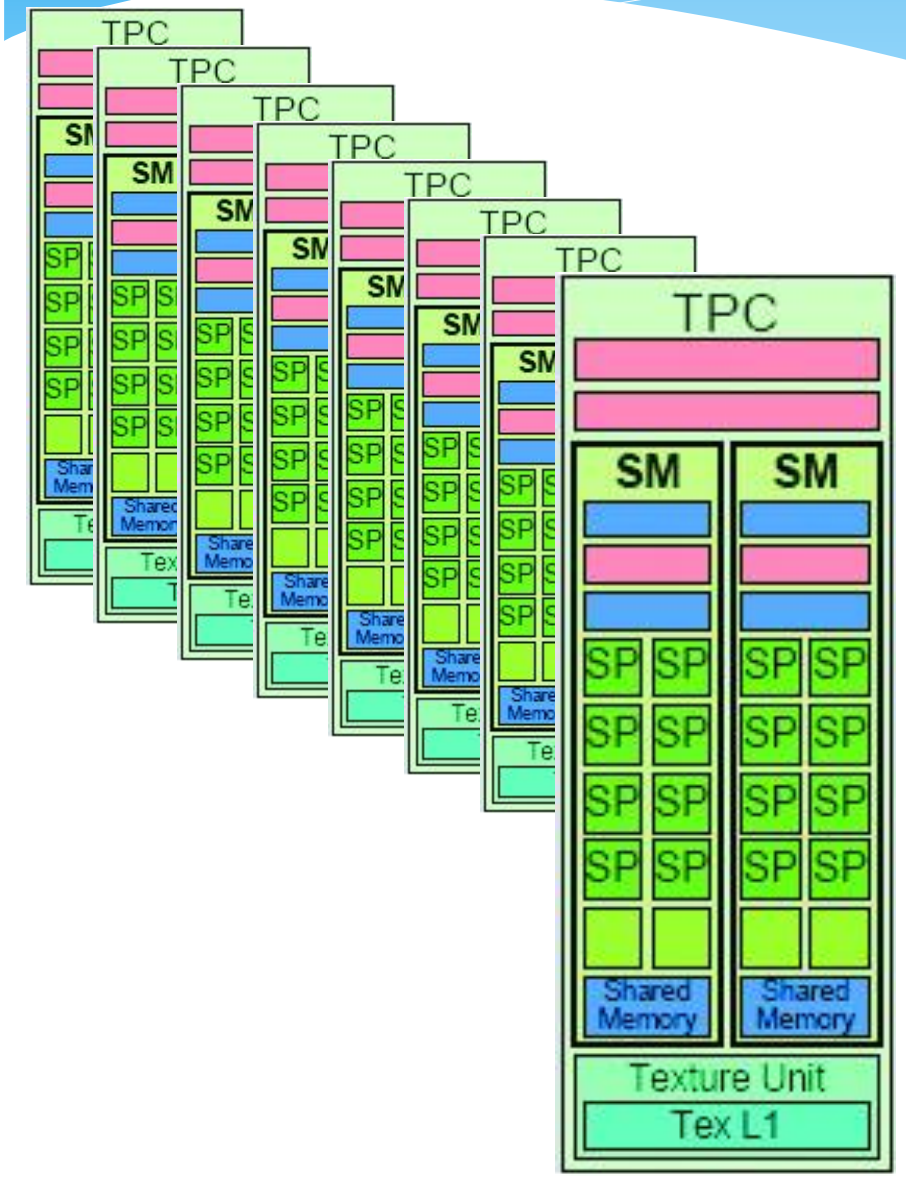
Fermi processor die



Memory access



Architecture of GPU (GT200)



* TPC - Texture Processor Cluster
* SM — Streaming Multiprocessor
  * Multi-threaded processor core
  * Fundamental processing unit for CUDA thread block
* SP — Streaming Processor
  * Scalar ALU for a single CUDA thread

	Number of SM
GeForce 8800 GTX	16
GeForce 8800 GTS	12
Tesla D870	2x16
Tesla S870	4x16
Tesla C1060, GT200, Tesla T10	30
Tesla S1070	4x30

Compute capatibility

* Compute Capability 1.0+
  * Asynchronous kernel execution
* Compute Capability 1.1+
  * Asynchronous copy engine (single engine). Property **asyncEngineCount**
  * Atomic operation
* Compute Capability 1.3+ (e.g. C1060)
  * Double precision arithmetic
* Compute Capability 2.0+ (e.g. C2050)
  * Parallel kernel execution on GPU (property **concurrentKernels**)
  * Two asynchronous copy engines (property **asyncEngineCount**)
* Compute Capability 3.0+ (e.g. K10)
  * Warp shuffle functions
* Compute Capability 3.5+ (e.g. K20)
  * Funnel shift
  * Dynamic parallelism

http://en.wikipedia.org/wiki/CUDA

Tesla C1060

1 TFlops



Tesla S1070

4 TFlops



DGX-A100



SYSTEM SPECIFICATIONS

	NVIDIA DGX A100 640GB
GPUs	**8x NVIDIA A100 80GB Tensor Core GPUs**
GPU Memory	**640GB total**
Performance	**5 petaFLOPS AI** **10 petaOPS INT8**
NVIDIA NVSwitches	**6**
System Power Usage	**6.5 kW max**
CPU	**Dual AMD Rome 7742, 128 cores total, 2.25 GHz (base), 3.4 GHz (max boost)**
System Memory	**2TB**

Useful links

* http://www.nvidia.com/object/cuda_home.html
* http://www.gputechconf.com/page/home.html
* http://docs.nvidia.com/
* https://youtu.be/PWcNlRI00jo - Keynote 2022

Questions